



Security Proof

- Full security proof $\Rightarrow$ all possible attacks of Eve.
- Alice: bit-string A ; Bob: bit-string B
- Eve: bit-string E the best guess she can make
- If QBER low then A, B more correlated than A, E or B, E : $I(A : B) > I(A : E)$
- Alice/Bob advantage in the final post-processing
- **Information Reconciliation (IR)**: Exchange information (error-correcting codes) to make $A = B$ (extra info leaked to Eve)
- **Privacy Amplification (PA)**: Distil shorter key completely secret from Eve (use universal hash functions to amplify privacy)



Realistic QKD

- Realistic systems have noise: $\text{QBER} \neq 0$ even if honest
- Cannot tell errors from noise Vs errors from eavesdropping
- QBER is used for:
 - Estimate correlation of Alice's raw bit-string A with Bob's B
 - Bound the max correlation that any adversary's bit string E can have with A
- If (A, B) "correlation" is higher than (A, E) then it is possible for Alice and Bob to distil an (identical) bit-string A totally secret from Eve (using IR & PA)
- The key-rate R , highest possible noise-tolerance and maximum distance possible all depend on the advantage $I(A : B) - I(A : E)$



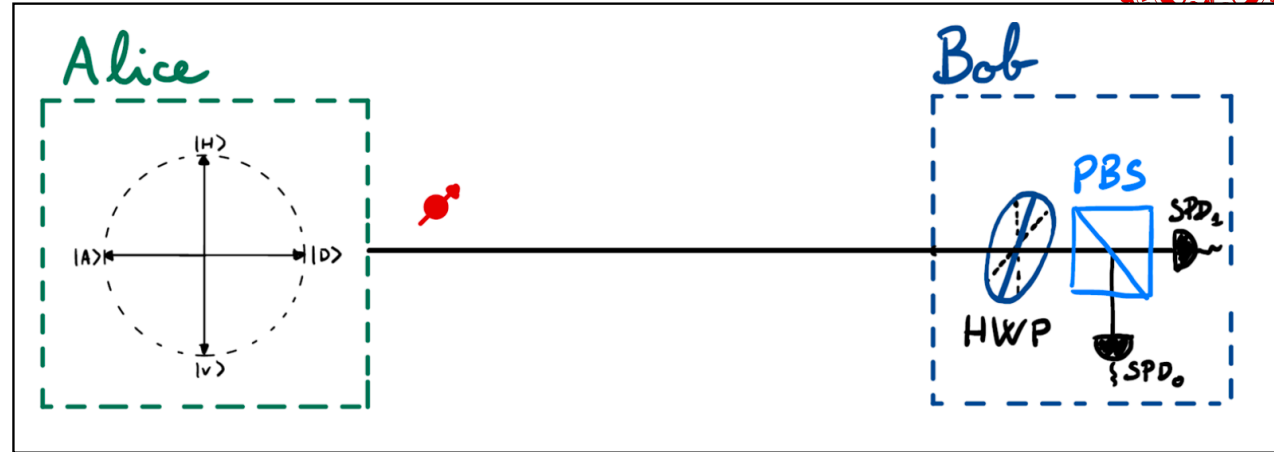
SKR of DV-QKD Protocol

- SKR: $R = I(A : B) - I(A : E)$
- Let the QBER be Q .
- Then $I(A : B) = 1 - h_2(Q)$ (Max mutual information of Binary symmetric channel)
- Max information leaked to Eve: $I(A : E) \leq h_2(Q)$
- $R = 1 - h_2(Q) - h_2(Q) = 1 - 2h_2(Q)$
- Adding sifting and Efficiency factors:
 - Only half of the bits are kept after sifting $q = 1/2$
 - Error Correction is not perfect: Inefficiency factor $f \geq 1$
- **Realistic SKR:** $R = q[1 - fh_2(Q) - h_2(Q)]$
- QBER Threshold for positive SKR: $1 - 2h_2(Q) = 0$ gives $Q \approx 0.11$

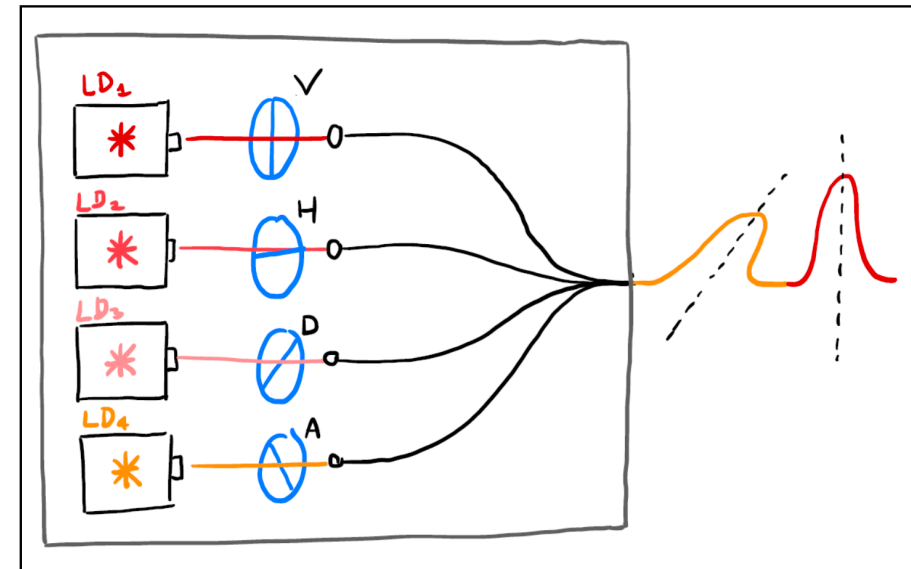
Polarisation Encoding BB84 QKD



- Alice randomly generates one of the 4 **Polarisation States**:
 - ✓ $|H\rangle$: Horizontal polarization (0°)
 - ✓ $|V\rangle$: Vertical polarization (90°),
 - ✓ $|D\rangle$: Diagonal polarization ($+45^\circ$)
 - ✓ $|A\rangle$: Anti-diagonal polarization (-45°)



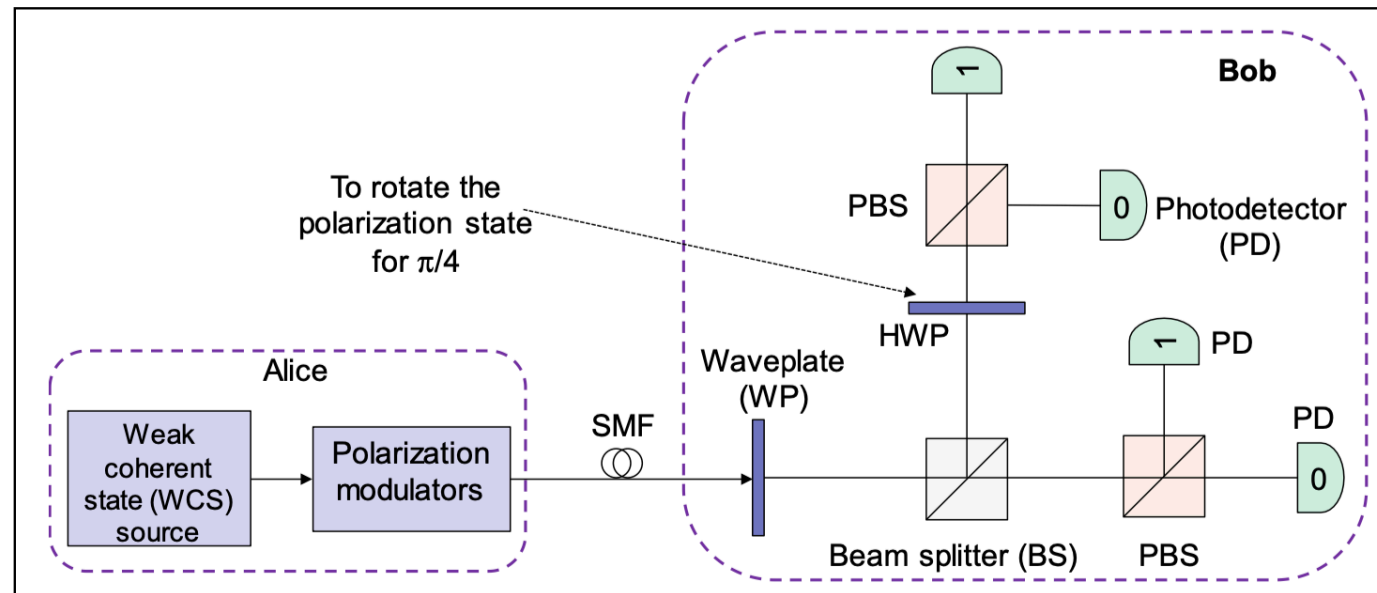
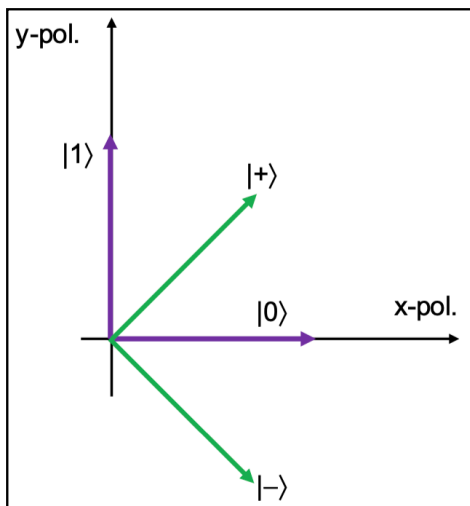
- **Half Wave Plate (HWP)**: Selects the measurement basis:
 - ✓ Rectilinear basis $\rightarrow$ HWP at one angle
 - ✓ Diagonal basis $\rightarrow$ HWP rotated by 22.5°
- **Polarizing Beam Splitter (PBS)**: Separates photons according to polarization into two spatial paths:
 - ✓ Horizontal photon polarisation: $\rightarrow$ exits one port
 - ✓ Vertical photon polarisation $\rightarrow$ exits the other port
- **Polarisation Encoding**: Used mostly in free space communication. In fiber requires an active tracking of the polarization



Polarization-Based BB84 QKD Protocol



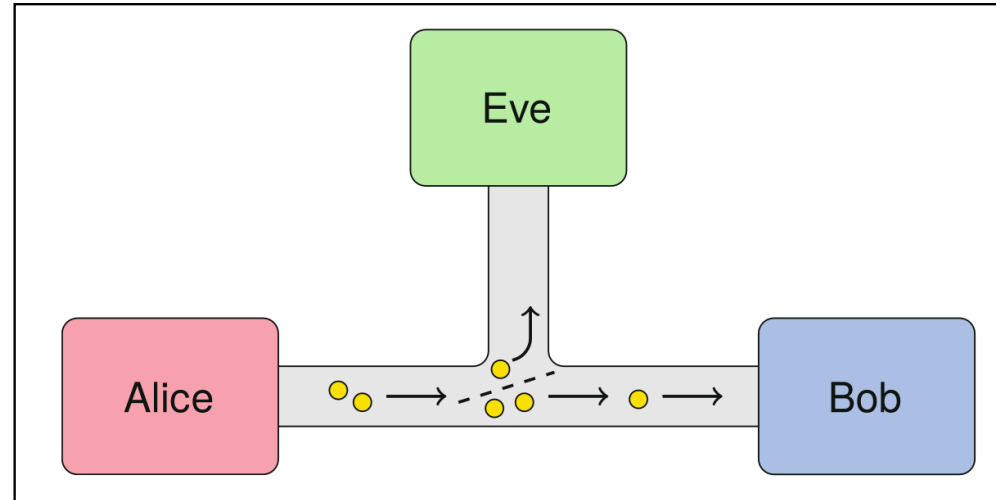
- Alice: 4-polarization states used
- Bob: Passive-basis-choice: A **50:50 beam splitter (BS)** randomly selects the measurement basis for each incoming photon
- Waveplate: Used for polarization compensation, and correcting fiber-induced polarization rotations.



Photon-Number-Splitting (PNS) Attack



- Ideal QKD uses single photons for qubits (e.g., polarization states).
- In practice, weak laser pulses often emit multiple photons.
- **PNS Attack:** Eve intercepts and keeps one photon from multi-photon pulses, waits for Alice's basis announcement, then measures stored photons in correct basis.
- This lets Eve gain perfect knowledge of the key without detection.
- Advanced BB84 QKD to overcome PNS Attack: **Decoy State QKD**.





Decoy State QKD

- **WCP Photon Distribution:** $P(n) = \frac{\mu^n e^{-\mu}}{n!}$
- Eve can perform **Quantum Non Demolition Measurement** to learn n without disturbing encoded qubit.
 - If $n = 1$: Then Eve blocks or forwards depending on attack strategy
 - If $n \geq 2$: Eve removes one photon and forwards the remaining photons to Bob
 - When Alice and Bob publicly announce their measurement bases, Eve measures the stored photon in correct basis and obtained the transmitted bits without introducing any detectable errors
- **Decoy State Method:** Alice varies the intensity of each transmitted pulse:
 - **Signal state:** intensity μ (used to generate the actual key)
 - **Decoy state(s):** intensity ν (used only for parameter estimation)
 - **Vacuum state** (optional but standard): intensity 0
- The crucial observation: A single-photon pulse is physically identical regardless of whether it originates from a signal state or decoy state.
- Therefore, Eve cannot distinguish whether a detected single photon belongs to a signal pulse or a decoy pulse.



Decoy-State BB84 Protocol

- **Step 1: State Preparation:** For every transmitted pulse, Alice randomly selects:
 - ✓ A pulse intensity (signal, weak decoy, or vacuum),
 - ✓ A basis (Z or X), and a bit value (0 or 1).
 - ✓ The bit is encoded into one of the BB84 polarization states.
- **Step 2: Quantum Transmission & Measurement:** Alice transmits the encoded WCP through the quantum channel. Bob independently chooses a measurement basis and records whether a detection event occurs.
- **Step 3: Basis Reconciliation:** Using authenticated channel, Alice & Bob reveal only their basis choices. Bits corresponding to different bases are discarded, producing the sifted key.
- **Step 4: Intensity Announcement:** After basis reconciliation, Alice announces which pulses were signal states and which were decoy states. Bob classifies his detection events according to the transmitted intensity.
- **Step 5: Parameter Estimation:** For each intensity level, Alice and Bob estimate the **gain**:
$$Q_{\mu} = \frac{\text{Detected pulses}}{\text{Transmitted pulses}} \text{ \& QBER: } E_{\mu} = \frac{\text{Number of erroneous bits}}{\text{Detected bits}}$$



Decoy-State BB84 SKR Analysis

- **Gain:** $Q_\mu = \sum_{n=0}^{\infty} P_\mu(n) Y_n$, where Y_n is the **yield**, i.e., the probability that Bob detects an n -photon pulse.
- Similarly, $E_\mu Q_\mu = \sum_{n=0}^{\infty} P_\mu(n) Y_n e_n$, where e_n is the error probability associated with an n -photon pulse.
- **Security Analysis:** The security of decoy-state QKD relies on the fact that Eve cannot distinguish whether a given n -photon pulse originates from a signal state or a decoy state.
- Consequently, the yield and error rate depend only on the photon number, $Y_n^{(\text{signal})} = Y_n^{(\text{decoy})}$ and $e_n^{(\text{signal})} = e_n^{(\text{decoy})}$
- **SKR:** $R = q \left[-Q_\mu f(E_\mu) H_2(E_\mu) + Q_1 (1 - H_2(e_1)) \right]$,
where , q is the sifting factor, Q_μ is the signal-state gain, , $Q_1 = P_\mu(1)Y_1$ is the single-photon gain,
- E_μ is the observed QBER, , e_1 is the single-photon error rate,
- $f(E_\mu)$ is the error-correction efficiency, $H_2(x)$ is the binary entropy function.
- **Step 6:** Error Correction and Privacy Amplification

Decoy-State BB84 Protocol Security



- Suppose Eve attempts a PNS attack by blocking all single-photon pulses and forwarding only multi-photon pulses.
- Decoy and signal states have different photon-number distributions due to their different mean photon numbers.
- A PNS attack **alters the expected detection statistics** (gain and QBER) of signal and decoy states.
- During **parameter estimation**, Alice and Bob compare the observed statistics with the theoretical Poisson predictions.
- Any inconsistency reveals Eve's presence, making the PNS attack detectable.
- **Result:** Decoy-state QKD effectively defeats PNS attacks while using practical weak coherent pulse sources.

Advantages of Decoy State QKD



- Provides unconditional security using inexpensive weak coherent laser sources.
- Effectively defeats PNS attacks.
- Significantly increases the achievable transmission distance.
- Produces higher secret key rates than standard weak-coherent BB84.
- Requires only an additional **intensity modulator**, making it easy to integrate into existing BB84 systems.
- Forms the basis of nearly all modern commercial discrete-variable QKD systems.